

10- پیوست شماره 1: فرآیند راه اندازی Apache Hadoop بر روی Ubuntu 24.04 LTS

Apache Hadoop v.3.3.5 on Java-11-openjdk and Ubuntu 24.04 LTS

1-System Update

```
sudo apt update
```

```
sudo apt upgrade -y
```

2-Install Java 11

```
sudo apt install openjdk-11-jdk -y
```

```
java -version
```

```
readlink -f $(which java)
```

3-Set Local Java Variables

```
nano ~/.bashrc
```

add below lines at the end of file:

```
export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64
```

```
export PATH=$PATH:$JAVA_HOME/bin
```

```
source ~/.bashrc
```

```
echo $JAVA_HOME
```

4-Download Apache Hadoop

```
wget https://downloads.apache.org/hadoop/common/hadoop-3.3.5/hadoop-3.3.5.tar.gz
```

```
https://archive.ito.gov.ir/
```

```
sudo tar -xvzf hadoop-3.3.5.tar.gz
```

```
sudo mv hadoop-3.3.5 /usr/local/Hadoop
```

```
ls /usr/local/Hadoop
```

5-Set Hadoop Local Variable

```
nano ~/.bashrc
```

```
export HADOOP_HOME=/usr/local/hadoop
```

```
export HADOOP_INSTALL=$HADOOP_HOME
```

```
export HADOOP_MAPRED_HOME=$HADOOP_HOME
export HADOOP_COMMON_HOME=$HADOOP_HOME
export HADOOP_HDFS_HOME=$HADOOP_HOME
export YARN_HOME=$HADOOP_HOME
export HADOOP_COMMON_LIB_NATIVE_DIR=$HADOOP_HOME/lib/native
export PATH=$PATH:$HADOOP_HOME/sbin:$HADOOP_HOME/bin
```

```
source ~/.bashrc
```

```
echo $HADOOP_HOME
```

6-Set Java at Hadoop

```
nano /usr/local/hadoop/etc/hadoop/hadoop-env.sh
```

```
export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64
```

7-Create User

```
sudo adduser Hadoop
```

```
sudo usermod -aG sudo Hadoop
```

```
sudo chown -R hadoop:hadoop /usr/local/Hadoop
```

```
sudo su - Hadoop
```

```
nano ~/.bashrc
```

```
export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64
export HADOOP_HOME=/usr/local/hadoop
export HADOOP_INSTALL=$HADOOP_HOME
export HADOOP_MAPRED_HOME=$HADOOP_HOME
export HADOOP_COMMON_HOME=$HADOOP_HOME
export HADOOP_HDFS_HOME=$HADOOP_HOME
export YARN_HOME=$HADOOP_HOME
export HADOOP_COMMON_LIB_NATIVE_DIR=$HADOOP_HOME/lib/native
export PATH=$PATH:$HADOOP_HOME/sbin:$HADOOP_HOME/bin
```

```
source ~/.bashrc
```

8-Install SSH

```
sudo apt install openssh-server -y  
sudo systemctl status ssh  
ssh-keygen -t rsa -P ""  
cat ~/.ssh/id_rsa.pub >> ~/.ssh/authorized_keys  
ssh localhost  
ssh-keygen -t rsa -P ""  
cat ~/.ssh/id_rsa.pub >> ~/.ssh/authorized_keys  
ssh localhost
```

9-Set Hadoop Configs:

Configs: cd /usr/local/hadoop/etc/Hadoop

core-site.xml

```
<configuration>  
  <property>  
    <name>fs.defaultFS</name>  
    <value>hdfs://localhost:9000</value>  
  </property>  
</configuration>
```

hdfs-site.xml

```
<configuration>  
  <property>  
    <name>dfs.replication</name>  
    <value>1</value>  
  </property>  
  <property>  
    <name>dfs.namenode.name.dir</name>  
    <value>file:///usr/local/hadoop/data/nn</value>  
  </property>  
  <property>
```

```
<name>dfs.datanode.data.dir</name>

<value>file:///usr/local/hadoop/data/dn</value>

</property>

</configuration>
```

mapred-site.xml

cp mapred-site.xml.template mapred-site.xml

```
<configuration>

<property>

  <name>mapreduce.framework.name</name>

  <value>yarn</value>

</property>

<property>

  <name>yarn.app.mapreduce.am.env</name>

  <value>HADOOP_MAPRED_HOME=/usr/local/hadoop</value>

</property>

<property>

  <name>mapreduce.map.env</name>

  <value>HADOOP_MAPRED_HOME=/usr/local/hadoop</value>

</property>

<property>

  <name>mapreduce.reduce.env</name>

  <value>HADOOP_MAPRED_HOME=/usr/local/hadoop</value>

</property>

</configuration>
```

yarn-site.xml

```
<configuration>

  <property>

    <name>yarn.nodemanager.aux-services</name>

    <value>mapreduce_shuffle</value>
```

```
</property>
```

```
</configuration>
```

10-Create Data Folders

```
sudo mkdir -p /usr/local/hadoop/data/nn
```

```
sudo mkdir -p /usr/local/hadoop/data/dn
```

```
sudo chown -R $USER:$USER /usr/local/hadoop/data
```

11-Hadoop Test

```
hadoop version
```

Important Paths:

Java: /usr/lib/jvm/java-11-openjdk-amd64

Hadoop: /usr/local/Hadoop

Hadoop Config: /usr/local/hadoop/etc/Hadoop

12-Run

```
start-dfs.sh
```

```
start-yarn.sh
```

```
jps
```

HDFS Web UI: <http://localhost:8088>

← → ↻ ⚠ Not secure 192.168.1.104:8088/cluster



Cluster

[About](#)
[Nodes](#)
[Node Labels](#)
[Applications](#)
[NEW](#)
[NEW SAVING](#)
[SUBMITTED](#)
[ACCEPTED](#)
[RUNNING](#)
[FINISHED](#)
[FAILED](#)
[KILLED](#)
[Scheduler](#)

Tools

Cluster Metrics

Apps Submitted	Apps Pending	Apps Running	Apps Completed
0	0	0	0

Cluster Nodes Metrics

Active Nodes	Decommissioning Nodes
1	0

Scheduler Metrics

Scheduler Type	Scheduling Resource Type
Capacity Scheduler	[memory-mb (unit=M), vcores]

Show 20 entries

ID	User	Name	Application Type	Application Tags	Queue	Application Priority	StartTime	Laun
Showing 0 to 0 of 0 entries								

YARN Web UI: <http://localhost:9870>

⚠ Not secure 192.168.1.104:9870/dfshealth.html#tab-overview

Hadoop

Overview

Datanodes

Datanode Volume Failures

Snapshot

Startup Progress

Utilities

Overview 'localhost:9000' (✓active)

Started:	Wed Mar 25 00:02:49 +0330 2026
Version:	3.3.5, r706d88266abcee09ed78fbaa0ad5f74d818ab0e9
Compiled:	Wed Mar 15 19:26:00 +0330 2023 by stevel from branch-3.3.5
Cluster ID:	CID-244e8289-088e-493a-8f91-e7b982624098
Block Pool ID:	BP-1115597152-127.0.1.1-1774382423380

Summary

Security is off.

Safemode is off.

1 files and directories, 0 blocks (0 replicated blocks, 0 erasure coded block groups) = 1 total filesystem object(s).

Heap Memory used 35.25 MB of 69.5 MB Heap Memory. Max Heap Memory is 477.56 MB.

Non Heap Memory used 51.03 MB of 54.25 MB Committed Non Heap Memory. Max Non Heap Memory is <unbounded>.

13-Built-in Word Count Example

```
hdfs dfs -mkdir /input
```

```
hdfs dfs -ls /
```

```
nano test.txt
```

```
hello hadoop
```

```
hello world
```

```
hello ubuntu
```

```
hadoop hdfs yarn
```

```
hdfs dfs -put test.txt /input
```

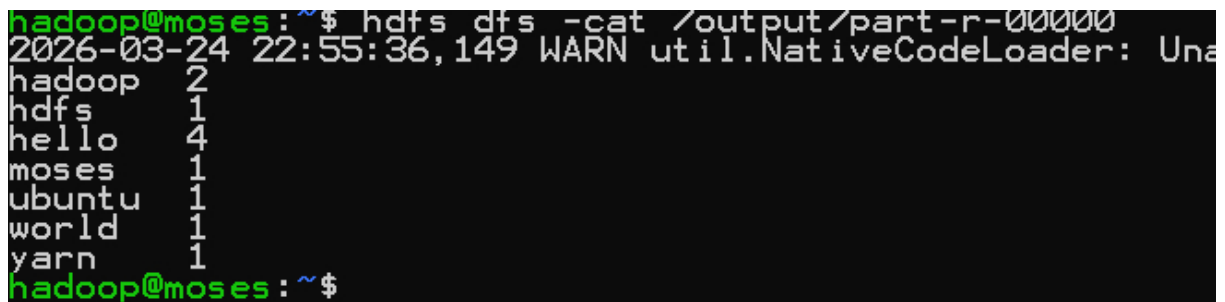
```
hdfs dfs -ls /input
```

```
hadoop jar \
```

```
$HADOOP_HOME/share/hadoop/mapreduce/hadoop-mapreduce-examples-3.3.5.jar \
```

```
wordcount /input/test.txt /output
```

```
hdfs dfs -cat /output/part-r-00000
```



```
hadoop@moses:~$ hdfs dfs -cat /output/part-r-00000
2026-03-24 22:55:36,149 WARN util.NativeCodeLoader: Unable to load native-hadoop library for optimization
hadoop      2
hdfs        1
hello       4
moses        1
ubuntu       1
world        1
yarn         1
hadoop@moses:~$
```

14-Linux Start-up

```
sudo nano /etc/systemd/system/hadoop-hdfs.service
```

```
[Unit]
```

```
Description=Hadoop HDFS
```

```
After=network.target
```

```
[Service]
```

```
User=hadoop
```

```
Type=forking
ExecStart=/usr/local/hadoop/sbin/start-dfs.sh
ExecStop=/usr/local/hadoop/sbin/stop-dfs.sh
RemainAfterExit=yes
[Install]
WantedBy=multi-user.target
```

sudo nano /etc/systemd/system/hadoop-yarn.service

```
[Unit]
Description=Hadoop YARN
After=network.target

[Service]
User=hadoop
Type=forking
ExecStart=/usr/local/hadoop/sbin/start-yarn.sh
ExecStop=/usr/local/hadoop/sbin/stop-yarn.sh
RemainAfterExit=yes
[Install]
WantedBy=multi-user.target
```

sudo systemctl daemon-reexec

sudo systemctl daemon-reload

sudo systemctl enable hadoop-hdfs

sudo systemctl enable hadoop-yarn

sudo systemctl start hadoop-hdfs

sudo systemctl start hadoop-yarn

sudo systemctl status hadoop-hdfs

sudo systemctl status hadoop-yarn

jps

sudo reboot


```
moses@moses:~$ sudo su
[sudo] password for moses:
root@moses:/home/moses# systemctl status hadoop-hdfs
' hadoop-hdfs.service - Hadoop HDFS
   Loaded: loaded (/etc/systemd/system/hadoop-hdfs.service; enabled; preset: enabled)
   Active: active (exited) since Tue 2026-03-24 23:14:40 UTC; 49s ago
   Process: 650 ExecStart=/usr/local/hadoop/sbin/start-dfs.sh (code=exited, status=0/SUCCESS)
   CPU: 13.091s

Mar 24 23:13:54 moses systemd[1]: Starting hadoop-hdfs.service - Hadoop HDFS...
Mar 24 23:13:59 moses start-dfs.sh[650]: Starting namenodes on [localhost]
Mar 24 23:14:10 moses start-dfs.sh[650]: Starting datanodes
Mar 24 23:14:16 moses start-dfs.sh[650]: Starting secondary namenodes [moses]
Mar 24 23:14:40 moses systemd[1]: Started hadoop-hdfs.service - Hadoop HDFS.
root@moses:/home/moses# systemctl status hadoop-yarn
' hadoop-yarn.service - Hadoop YARN
   Loaded: loaded (/etc/systemd/system/hadoop-yarn.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-03-24 23:14:12 UTC; 1min 22s ago
   Process: 651 ExecStart=/usr/local/hadoop/sbin/start-yarn.sh (code=exited, status=0/SUCCESS)
   Main PID: 953 (java)
   Tasks: 219 (limit: 2268)
   Memory: 250.5M (peak: 303.8M)
   CPU: 11.977s
   CGroup: /system.slice/hadoop-yarn.service
```